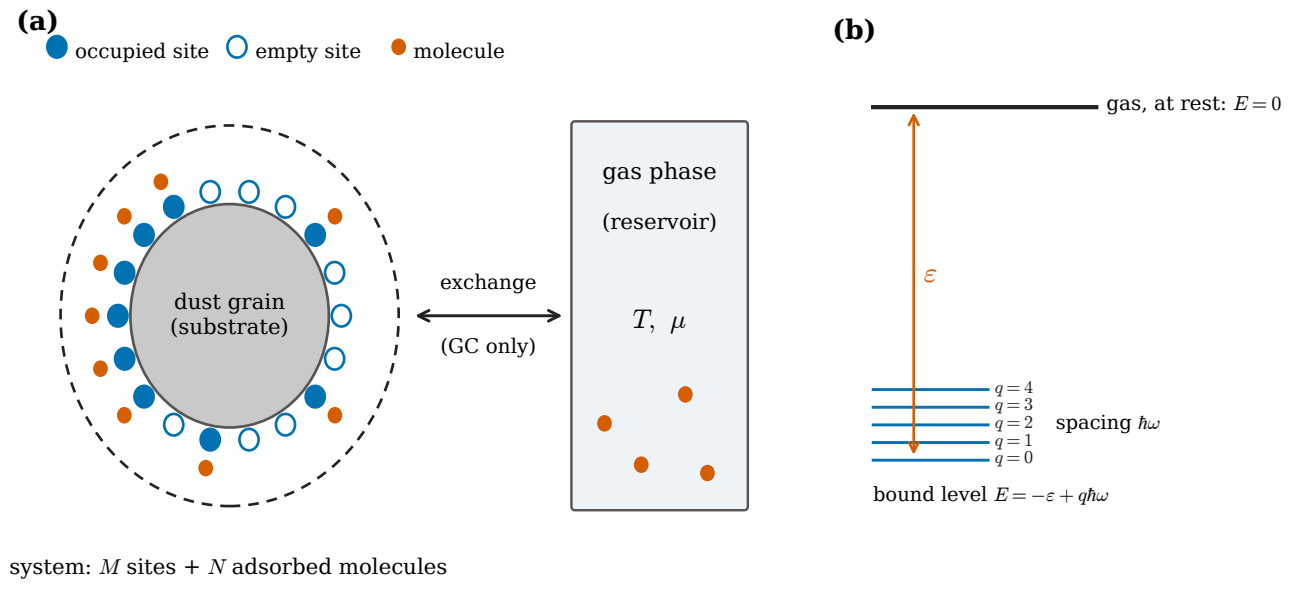
(a)
occupied site
empty site
molecule
dust grain
(substrate)
exchange
(GC only)
gas phase
(reservoir)
$T,\ \mu$
(b)
gas, at rest: $E = 0$
ε
$q = 4$
$q = 3$
$q = 2$
$q = 1$
$q = 0$
spacing $\hbar\omega$
bound level $E = -\varepsilon + q\hbar\omega$
system: M sites + N adsorbed molecules